Anomalies Système : Mémoire Réservée au Matériel, Popups 0x00 et Artefacts iGPU

1. Contexte et Description Technique

Résolution des dysfonctionnements logiciels sous Windows liés à l'allocation mémoire, aux fuites de mémoire (Out of Memory) et à la corruption graphique sur GPU intégré (iGPU).

2. Symptômes Observés sur la Machine

Le Gestionnaire des tâches affiche 8 Go 'Matériel réservé' sur 16 Go installés, popups d'erreur d'application 'The instruction at 0x... referenced memory could not be read', ou damiers colorés à l'écran (iGPU).

3. Protocole de Diagnostic et Résolution

1. Pour la mémoire 'Matériel réservé' excessive : vérifier dans `msconfig` > Démarrer > Options avancées que la case 'Mémoire maximale' est bien décochée, puis réinsérer les barrettes.

2. Pour les artefacts visuels sur processeur avec graphiques intégrés (APU Ryzen G / Intel UHD), la VRAM étant partagée avec la RAM système, remplacer ou abaisser la fréquence de la RAM corrige immédiatement l'affichage.

4. Articles et Retours d'Expérience Extraits du Web

Extrait de [Windows was reserving my RAM. Here’s how I got my usable ...] :
After upgrading my PC’s RAM, I expected smoother multitasking, faster game loading, and plenty of memory for dozens of browser tabs. Instead, Windows still felt sluggish. When I opened Task Manager, I discovered that several gigabytes of RAM were marked as Hardware Reserved. At first, I assumed Windows was simply using that memory for background processes. It wasn’t. That RAM wasn’t available to Windows at all. If you’re seeing a similar issue, here’s exactly how I got reserved RAM back on Windows 11 and increased usable RAM. How I found Windows wasn’t using all my RAM I first noticed something was off when my PC started slowing down while multitasking. I knew that I had sufficient RAM to handle whatever I was doing, so I decided to go into Task Manager and check the Memory tab to investigate further. It showed that Windows had less usable memory than the amount installed in my system. For example, I had 8GB of RAM installed, but only around 5.9GB was available. The rest was listed as Hardware Reserved. To confirm it wasn’t just a reporting issue, I checked Settings → System → About and the System Information app. Both showed that all of my RAM was installed, but Windows still wasn...

Extrait de [Hardware Reserved Memory Fix (Windows RAM Limits) - LaptopJudge] :
Hardware Reserved Memory Fix (Windows RAM Limits) Hardware-reserved memory is RAM that Windows cannot use because firmware or devices have claimed it. First compare installed memory with Windows-visible memory, then check for a Windows boot limit. If no limit exists, inspect firmware settings, graphics memory, and memory modules in a careful order. Change one setting at a time, and verify the result after each restart. What if Task Manager shows 16 GB installed but only 8 GB usable, just as a video call starts to stutter? It is tempting to end background tasks or change Windows settings at random. But hardware-reserved memory is not a process you can close, and the right fix depends on where the missing memory is being held. I start by comparing measurements, not guessing from a single screen. Windows may reserve some RAM for devices, and integrated graphics can use a share of system memory. A large, unexpected gap may also point to a boot setting, firmware issue, or poorly seated memory module. The steps below help distinguish these causes without treating normal reservations as faults. Measure the memory gap Hardware-reserved memory is installed RAM that firmware or hardware sets...

Sources Web Associées :

• Windows was reserving my RAM. Here’s how I got my usable ... (https://www.igeeksblog.com/fix-hardware-reserved-ram-windows/)

• Hardware Reserved Memory Fix (Windows RAM Limits) - LaptopJudge (https://laptopjudge.com/hardware-reserved-memory-fix/)

Catalogue Visuel : 10 Photos Réelles (10_os_memory_popups_artifacts)

Ces 10 cas visuels issus du dataset illustrent les motifs caractéristiques de cette classe d'erreur :

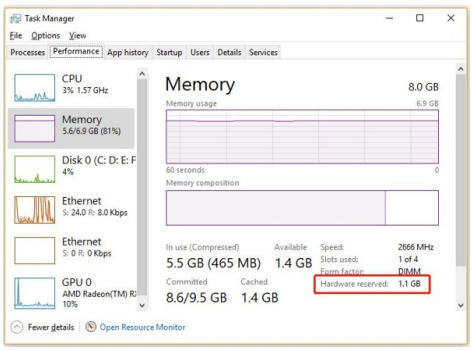


Figure 01 — 10_os_memory_popups_artifacts (Échantillon #1)



Figure 02 — 10_os_memory_popups_artifacts (Échantillon #2)

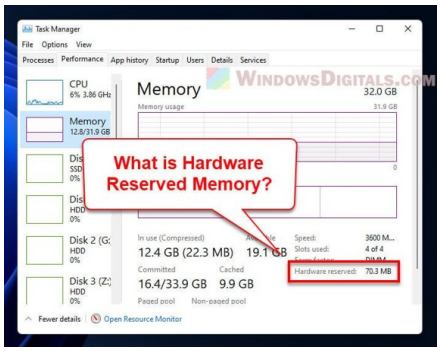


Figure 03 — 10_os_memory_popups_artifacts (Échantillon #3)



Figure 04 — 10_os_memory_popups_artifacts (Échantillon #4)

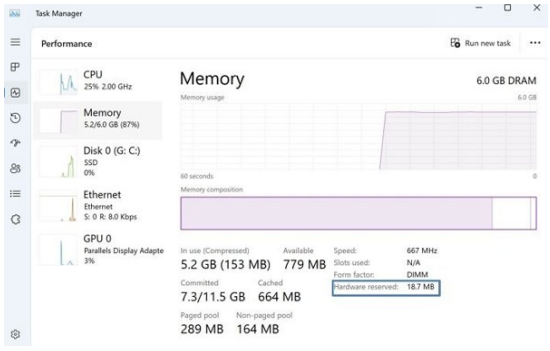


Figure 05 — 10_os_memory_popups_artifacts (Échantillon #5)

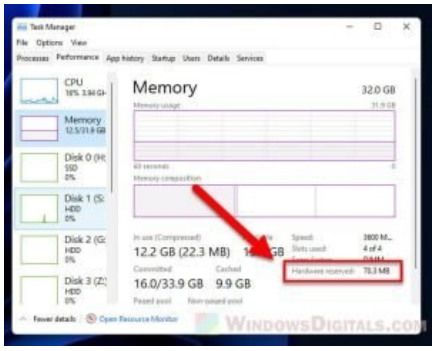


Figure 06 — 10_os_memory_popups_artifacts (Échantillon #6)

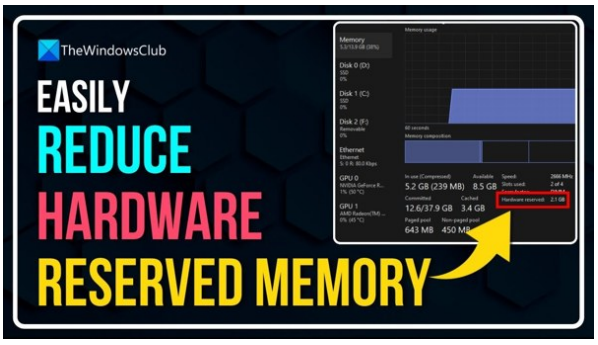


Figure 07 — 10_os_memory_popups_artifacts (Échantillon #7)

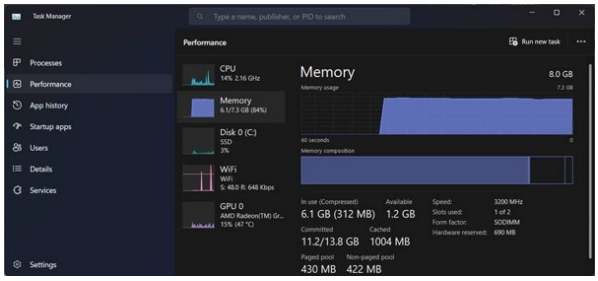


Figure 08 — 10_os_memory_popups_artifacts (Échantillon #8)

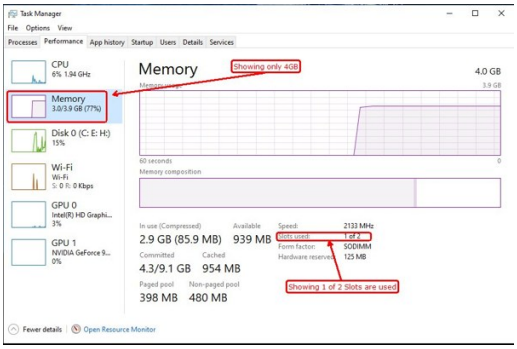


Figure 09 — 10_os_memory_popups_artifacts (Échantillon #9)

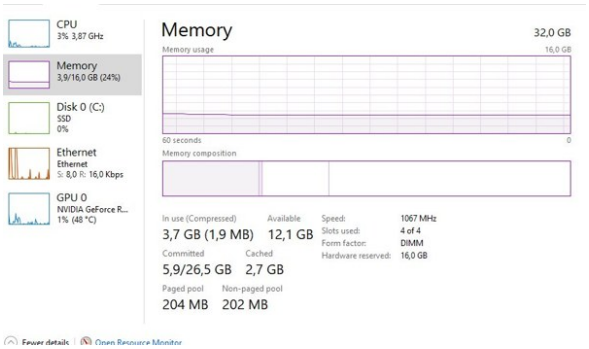


Figure 10 — 10_os_memory_popups_artifacts (Échantillon #10)